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TITLE PAGE

- **Problem Statement ID - SIH26165**
- **Problem Statement Title - AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - ____**
- **Team Name - Signal-0**

Working prototype: https://sanket-frontend.onrender.com

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SANKET — From Incident Narratives to Early Safety Warnings

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SANKET analyzes safety / near-miss reports using AI to detect potential Serious Injury & Fatality (SIF) precursors.

Identifies

- Hazard & Life-Saving Rule
- Control status
- Severity (1–5)
- SIF Precursor: Yes / No
- Evidence & reasoning

Innovation

- LLM + ML hybrid approach
- Explainable risk assessment
- One-Change severity rule
- Prioritized safety-review queue

HOW A REPORT IS JUDGED

Gate	Question	Possible answers
1 Hazard	Is a high-energy hazard present?	yes · no · insufficient information
2 Control	Was a safety barrier doing its job?	absent · failed · present · unclear
3 Severity	How bad could it realistically have been?	1 to 5

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TECHNICAL APPROACH

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Tech Stack

Layer	Technology	Purpose
Frontend	React	worker + admin screens
Backend	FastAPI (Python)	REST API, validation
AI model	Groq GPT-OSS-20B	reads the report
Fallback	TF-IDF + LogReg	answers if AI is down
Database	Supabase (PostgreSQL)	reports + predictions
Cache	SQLite	on-instance, repeat calls

Flow

Report → AI / NLP → Hazard → LSR → Control → Severity → SIF Decision → Risk Priority

SIF Rule

Hazard = YES

+ Control = Absent / Failed

+ Severity ≥ 4

→ SIF

All three, or it is not flagged. The flag is computed from the gates, never typed.

SYSTEM ARCHITECTURE

CLIENT		API		MODELS		DATA
React SPA	→	**FastAPI on Render**	→	**Groq GPT-OSS-20B**	→	**Supabase PostgreSQL**
worker screen · admin triage		REST · Pydantic validation · CORS		with local TF-IDF fallback		reports · predictions · labels

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FEASIBILITY AND VIABILITY

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Feasible because

- Uses existing incident reports
- No specialized hardware
- API-based & scalable
- LLM + local fallback
- Human-in-the-loop

CHALLENGES → SOLUTIONS

Challenge	Solution
Unstructured text	LLM reads it against a written rubric
Ambiguous controls	4-state classification, silence ≠ absent
API failure or rate limit	Retry inside a deadline, then local model
Domain variation	Tested on 30 real OSHA reports - F1 0.118, an honest gap
Model version changes	Dashboard scopes to one version at a time

HUMAN-IN-THE-LOOP — the system ranks, a person decides

Report arrives worker types or speaks it → **AI classifies** three gates + reasoning → **Queue reorders** highest risk first → **Officer reviews** reads the reasoning → **Officer decides** dispatch or archive

SANKET never closes a report. It only changes the order they are read in.

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IMPACT AND BENEFITS

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Potential impact on target audience

- Early detection of SIF precursors in OIL safety reports
- Prioritized review of high-risk incidents
- Faster identification of recurring hazards across sites / activities

Benefits

- Safety: proactive intervention before serious incidents
- Operational: reduces manual screening effort
- Economic: prevents costly injuries and downtime
- Scalable: handles large volumes of reports

MEASURED ON OUR PROTOTYPE DATA

What	Value	Meaning
Labelled corpus	180 reports	150 written for the prototype + 30 real OSHA narratives
Human reviews	360	every report read by two people against one written rubric
Agreed gold labels	173	7 reports still being adjudicated
Reviewer agreement	97% (kappa 0.947)	on a 38-report sample reviewed independently

Safety Reports → SIF Precursors → Risk Prioritization → Early Action → Safer Operations

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RESEARCH AND REFERENCES

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Research:

IOGP Life-Saving Rules • DEKRA SIF Framework • EEI SIF Model

Data:

150 synthetic safety reports + 30 OSHA severe-injury reports · 173 human labels

Prototype Evaluation:

Measure	Result	How it was measured
Human agreement ceiling	97% · kappa 0.947	two reviewers, working separately, n=38
TF-IDF baseline (F1)	0.754 ± 0.077	5-fold cross-validation, n=114
LLM classifier (F1)	being re-measured	previous figure withdrawn — see note below

Why the LLM figure is withdrawn

We found one of our held-out test reports written into the AI prompt, so the old score was not a fair test. We pulled the number rather than ship it.

What the ceiling means

Two people applying the same rubric to the same reports agreed 97% of the time. No classifier scored against those labels can honestly claim to be more consistent than the people who made them.

Live prototype: https://sanket-frontend.onrender.com **GitHub: SANKET — AI-powered SIF precursor detection**